MUHAMMAD ARBAB TAHIR

GIS Developer | Backend Developer | Geospatial Data Specialist

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PROFESSIONAL SUMMARY

GIS Developer and Software Development professional with experience in geospatial application development, backend architecture, and web technologies. Skilled in Node.js, Express.js, PostgreSQL/PostGIS, GeoServer, SQL, and JavaScript, with practical experience in developing web platforms, managing spatial databases, building interactive maps, and automating data workflows. Experienced in technical coordination, data integration, and delivering scalable digital solutions.

TECHNICAL SKILLS

Database Management: PostgreSQL, PostGIS, SQL
Backend Development: Node.js, Express.js
Web Technologies: HTML, CSS, JavaScript
GIS & Web Mapping: QGIS, ArcMap, GeoServer, Leaflet, GeoJSON
Remote Sensing & Climate: ENVI, Landsat, MODIS, NDVI, GrADS, ERA Reanalysis, RClimDex
Programming & Data Analysis: Python, R, MATLAB, C++
Scripting & Automation: PowerShell, Windows Batch, FFmpeg, Ghostscript, Git/GitHub
Visualization & Analysis: Excel Dashboards, GNUplot
Other: Project Management, Photogrammetry, NeRF
Soft Skills: Communication, Decision Making, Observation, Multi-tasking

PROFESSIONAL EXPERIENCE

Assistant Manager (GIS Developer) **Sep 2026 – Present**

National Disaster Management Authority (NDMA)

- Manage and develop GIS-based web portals for disaster management.
- Develop backend services using Node.js and Express.js.
- Manage PostgreSQL/PostGIS databases and GeoServer services.
- Create interactive maps and geospatial visualizations.
- Coordinate with partner agencies and NGOs for geospatial data integration and publishing.

Developer **Apr 2025 – Jun 2026**

Umar Wyne Official

- Designed, developed, and maintained web platforms, including backend architecture and data management modules.
- Standardized client data structures and centralized records to improve data organization and accuracy.
- Developed workflow solutions for profile tracking, activity histories, cross-functional coordination, and secure transaction logging.
- Optimized backend processes and operational workflows to improve system efficiency and data integrity.
- Translated functional requirements into scalable development solutions, enhancing platform functionality and user experience.

Social Media Specialist **Aug 2023 – Jun 2026**

Pakistan Television Corporation (PTV)

- Managed large datasets and digital information systems.
- Prepared analytical reports and presentations.
- Maintained databases and secure digital records.
- Coordinated with technical teams.
- Performed data analysis and reporting using Excel.

Intern **2022**

Pakistan Meteorological Department (PMD)

- Analyzed forecast models (ECMWF, ICON, GFS) against observed data.
- Prepared comparative reports to assess model accuracy for meteorological prediction.

Home Tutor **2022 – Present**
Freelance

- Deliver science and technical subject tutoring to students from middle school to BS level.
- Assist with assignments and exam preparation.

PROJECTS

Source code and documentation for most projects: github.com/marbab-boby

- **US Wildfires Web GIS (2005–2020)** | *QGIS, GeoServer, Leaflet, GeoJSON, MODIS NDVI*
  Mapped 4,100+ US wildfires as weighted heatmaps alongside MODIS NDVI and published them three ways: QGIS, a GeoServer web map and a live Leaflet/GeoJSON web app with a year slider.
- **Vegetation Change in Islamabad, 1990 vs 2020** | *R, Landsat 5/8, NDVI*
  Clipped and masked Landsat imagery to the ICT boundary, computed NDVI and difference maps, and quantified vegetated area growing from 203 km² to 624 km².
- **Climate Extremes at Kakul Station (1960–2013)** | *R, RClimDex, ETCCDI indices*
  Computed 32 ETCCDI climate-extreme indices from 54 years of daily station records and tested their trends for significance, assessing implications for local agriculture.
- **Windows Automation Toolkit** | *PowerShell, Batch, FFmpeg, Ghostscript, Python*
  Built and open-sourced (MIT) 8 no-install utilities for office workflows: batch rename with undo, duplicate file detection (SHA-256), Excel folder reports, file-size reduction, photo-to-video slideshows, PowerPoint slide and media export, and script-to-voice-over with subtitles.
- **Comparison of 3D Photogrammetry & NeRF Models (Final Year Project)** | *Photogrammetry, NeRF*
  Compared 3D reconstruction quality and performance of both approaches for AR/VR navigation.
- **South Asian Monsoon Circulation: Summer vs Winter** | *GrADS, ERA reanalysis*
  Produced temperature, wind and geopotential-height maps at 1000 and 500 hPa to analyse seasonal monsoon reversal and the Tibetan High.
- **Vector Calculus in MATLAB** | *MATLAB, Symbolic Math*
  Wrote scripts that compute and visualise gradient, divergence and curl of vector fields, including an electric-field model of a charged spherical shell.
- **Property Management System** | *C++*
  Created a data visualization and storage tool.
- **Space Policy Proposal for Pakistan** | *Policy research*
  Researched and drafted a strategic space policy framework.

EDUCATION

Bachelor of Science in Space Sciences | CGPA 3.37 **2019 – 2023**
Institute of Space Technology, Islamabad

Intermediate (Pre-Engineering) **2017 – 2019**
Punjab Group of Colleges, Abdul Hakim

Matriculation (Science) **2015 – 2017**
Govt. High School, Abdul Hakim

CERTIFICATIONS & TRAINING

- Excel Dashboard for Beginners (2025)
- Introduction to Spark AR – Meta (2025)
- Google Soft Skills Program (2024)
- Introduction to Data Science (2024)
- Image Processing Onramp – MathWorks (2023)

ACHIEVEMENTS

- Appreciation Letter – Ministry of Information & Broadcasting, Govt. of Pakistan (2024)
- Best Participation Award – 1st International Space Science & Payload Competition (2022)
- Bronze Honor – International Youth Math Challenge (2021)